

Exp.No. 2: Implementing TDMA Slot Allocation in GSM.

CODE IMPLEMENTATION :

```
gsm_tdma_simulation.m x
/MATLAB Drive/gsm_tdma_simulation.m
10
17 fprintf('Time Slot Duration = %.4f ms\n', slot_duration);
18 fprintf('Frame Utilization = %.2f %%\n', frame_utilization);
19 fprintf('Number of Allocated Time Slots = %d\n', allocated_slots);
20
21 figure;
22
23 subplot(1,3,1);
24 bar(slot_duration);
25 grid on;
26 title('Time Slot Duration');
27 ylabel('Time (ms)');
28 xticklabels({'Slot Duration'});
29
30 subplot(1,3,2);
31 bar(frame_utilization);
32 grid on;
33 title('Frame Utilization');
34 ylabel('Utilization (%)');
35 xticklabels({'Utilization'});
36
37 subplot(1,3,3);
38 bar([allocated_slots total_slots-allocated_slots]);
39 grid on;
40 title('Allocated vs Unallocated Slots');
41 ylabel('Number of Slots');
42 xticklabels({'Allocated','Unallocated'});

Command Window

New to MATLAB? See resources for Getting Started.

Time Slot Duration = 0.5769 ms
Frame Utilization = 75.00 %
Number of Allocated Time Slots = 6
>>
```

OUTPUT:

